Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Position

Meaning in my words How would you explain position to a friend?	**How can you tell?** The doorway is zero and east is positive. Where is a cart at −3 m?
Example or sketch How would you mark a cart 2 m east of the doorway on a number line?	**Non-example and why** “I walked 7 m.” Do we know your position now?

Distance

Meaning in my words How would you explain distance to a friend?	**How can you tell?** A cart goes 3 m right and 2 m left. How much ground do its wheels cover?
Example or sketch Draw a trip that covers 6 m and finishes at its starting point.	**Non-example and why** A partner writes −5 m for distance. What does the minus sign tell you?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Displacement

Meaning in my words How would you explain displacement to a friend?	**How can you tell?** Which two marks tell you the displacement even if the cart turns around?
Example or sketch Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement?	**Non-example and why** Someone adds both legs and gets 6 m. What did they find?

Scalar / vector

Meaning in my words How would you explain the difference between a scalar and a vector?	**How can you tell?** Which tells you which way to draw an arrow: 5 m or 5 m west?
Example or sketch Give a scalar and a vector for a 4 m walk straight east.	**Non-example and why** Two students each walk 4 m, one east and one west. Are their displacements equal?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Put start, turn, and finish on a number line. Label distance on the path and displacement on the shortcut.

Choose zero, direction, and scale.



My marks mean

Send the rover home

02

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Tell distance traveled and change in position apart.

CHOOSE ONE JOB TOGETHER

ROUTE ENGINEER ☐

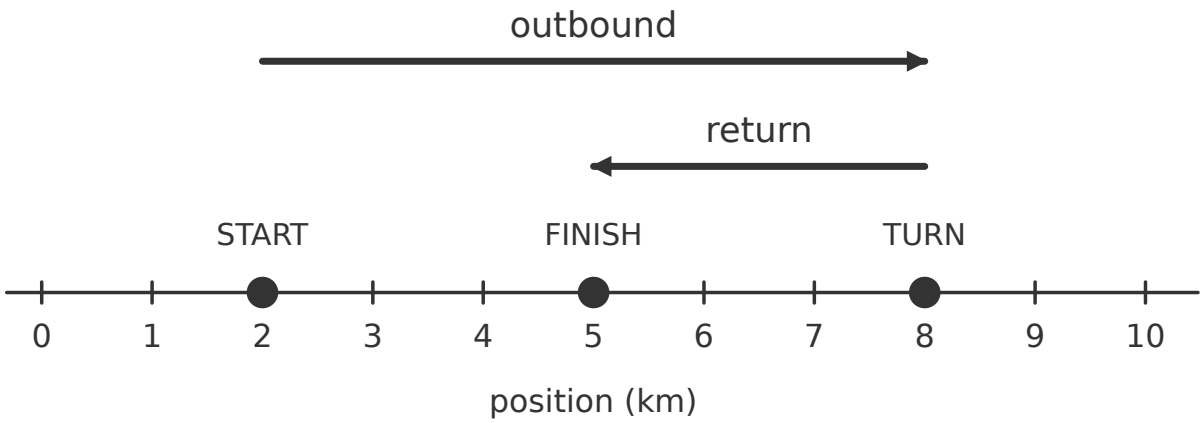
Mark both parts of the route and draw one start-to-finish arrow. Report the trip length and change in position.

RADIO DISPATCHER ☐

Guide a new driver to the finish. Give the trip length, change in position, and final location.

Point to a clue. Keep your first marks; add the update using the color key.

A survey rover tracks XJ's model from **+2** to **+8** to **+5 km**. Dispatch needs both mileage and final location.



01 OUR FIRST CALL

What numbers should dispatch use? Say what each number measures.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new location instruction. What should dispatch tell the driver now?

Switch speakers. If right becomes negative, which numbers change sign? Which quantities stay the same?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks